



Department of Statistics
and Data Science
Faculty of Science

Tutorial 10

Logistic Regression

ST3131 AY25/26 Sem2

2026 Apr. 15

1 What is logistic regression?

Definition 1. Denote by regressor $\mathbf{x}_i = (1, x_{i1}, \dots, x_{ik})$, the coefficient $\boldsymbol{\beta} = (\beta_0, \dots, \beta_k)$, and the response y_i .

- (Linear Regression) If y_i is continuous, then we assume

$$y_i = \mathbf{x}_i' \boldsymbol{\beta} + \epsilon_i \Rightarrow y_i \sim N(\mathbf{x}_i' \boldsymbol{\beta}, \sigma^2).$$

- (Logistic Regression) If y_i only takes two possible values $\{0, 1\}$, then we assume

$$p_i = \frac{\exp(\mathbf{x}_i' \boldsymbol{\beta})}{1 + \exp(\mathbf{x}_i' \boldsymbol{\beta})} \Rightarrow y_i \sim \text{Ber}(p_i)$$

- Compute an estimate $\hat{\boldsymbol{\beta}}$ for coefficients
- Test for individual $H_0 : \beta_j = 0$
- Test for the overall $H_0 : \text{some } \beta's = 0$
- Given new $\mathbf{x}$, predict its corresponding $\hat{y} = 0/1$

- 1 What is logistic regression? $p_i = \frac{\exp(\mathbf{x}_i' \boldsymbol{\beta})}{1 + \exp(\mathbf{x}_i' \boldsymbol{\beta})}$
- 2 Compute an estimate $\hat{\boldsymbol{\beta}}$
- 3 Test for individual $H_0 : \beta_j = 0$
- 4 Test for the overall $H_0 : \text{some } \beta'_s = 0$
- 5 Given new $\mathbf{x}$, predict its corresponding $\hat{y} = 0/1$

2 Compute an estimate $\hat{\beta}$

$$p_i = \frac{\exp(\mathbf{x}_i' \boldsymbol{\beta})}{1 + \exp(\mathbf{x}_i' \boldsymbol{\beta})}$$

Variable	Description
Survived	survival status: 1 = Yes, 0 = No
Class ($\beta_1, \beta_2, \beta_3$)	passenger's class: 1st, 2nd, 3rd and Crew
Sex (β_4)	gender of the passenger: Male, Female
Age (β_5)	age of the passenger: Child, Adult

Q1. Fit logistic regression model.

```
data = read.csv("titanic.csv")
data$y = ifelse(data$Survived=="1", 1, 0)
M = glm(y ~ Class + Sex + Age ,
        data = data,
        family = binomial)
summary(M)
```

Call: glm(formula = y ~ Class + Sex + Age)

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	2.0438	0.1679	12.171	< 2e-16	***
Class2nd	-1.0181	0.1960	-5.194	2.05e-07	***
Class3rd	-1.7778	0.1716	-10.362	< 2e-16	***
ClassCrew	-0.8577	0.1573	-5.451	5.00e-08	***
SexMale	-2.4201	0.1404	-17.236	< 2e-16	***
AgeChild	1.0615	0.2440	4.350	1.36e-05	***

Null deviance: 2769.5 on 2200 degrees of freedom

Residual deviance: 2210.1 on 2195 degrees of freedom

AIC: 2222.1

2 Compute an estimate $\hat{\beta}$

Q1. Fit logistic regression model.

$$p_i = \frac{\exp(\mathbf{x}_i' \boldsymbol{\beta})}{1 + \exp(\mathbf{x}_i' \boldsymbol{\beta})}$$

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$$\hat{p} = \frac{\exp(2 - x_1 - 1.7x_2 - 0.8x_3 - 2.4x_4 + x_5)}{1 + \exp(2 - x_1 - 1.7x_2 - 0.8x_3 - 2.4x_4 + x_5)}$$

2 Compute an estimate $\hat{\beta}$

Q2. Report and interpret $\hat{\beta}_4$

Variable	Description
Survived	survival status: 1 = Yes, 0 = No
Class ($\beta_1, \beta_2, \beta_3$)	passenger's class: 1st, 2nd, 3rd and Crew
Sex (β_4)	gender of the passenger: Male, Female
Age (β_5)	age of the passenger: Child, Adult

$$\hat{p} = \frac{\exp(\star - 2.4x_4)}{1 + \exp(\star - 2.4x_4)}$$

$$\hat{p} = \frac{\exp(2 - x_1 - 1.7x_2 - 0.8x_3 - 2.4x_4 + x_5)}{1 + \exp(2 - x_1 - 1.7x_2 - 0.8x_3 - 2.4x_4 + x_5)}$$

$$\hat{p}_{\text{male}} = \frac{\exp(\star - 2.4)}{1 + \exp(\star - 2.4)} \Rightarrow \text{odds(male)} := \frac{\hat{p}_{\text{male}}}{1 - \hat{p}_{\text{male}}} = \exp(\star - 2.4)$$

$$\hat{p}_{\text{female}} = \frac{\exp(\star)}{1 + \exp(\star)} \Rightarrow \text{odds(female)} := \frac{\hat{p}_{\text{female}}}{1 - \hat{p}_{\text{female}}} = \exp(\star)$$

$$\frac{\text{odds(male)}}{\text{odds(female)}} = \frac{\exp(\star - 2.4)}{\exp(\star)} = \cancel{e^{-2.4}}^{e^{\hat{\beta}_4}} \approx 0.09.$$

- 1 What is logistic regression? $p_i = \frac{\exp(\mathbf{x}_i' \boldsymbol{\beta})}{1 + \exp(\mathbf{x}_i' \boldsymbol{\beta})}$
- 2 Compute an estimate $\hat{\boldsymbol{\beta}}$
- 3 Test for individual $H_0 : \beta_j = 0$
- 4 Test for the overall $H_0 : \text{some } \beta'_s = 0$
- 5 Given new $\mathbf{x}$, predict its corresponding $\hat{y} = 0/1$

```
M = glm(y ~ Class + Sex + Age ,  
        data = data,  
        family = binomial)
```

3 Test for individual $H_0 : \beta_j = 0$

Q3. Perform Wald test for Sex at level 0.05

```
M = glm(y ~ Class + Sex + Age ,  
        data = data,  
        family = binomial)
```

$$H_0 : \beta_{\text{SexMale}} = 0$$

Test statistic:

$$\frac{\hat{\beta}_{\text{SexMale}}}{\text{se}(\hat{\beta}_{\text{SexMale}})} \stackrel{H_0}{\sim} N(0, 1)$$

$$\frac{-2.4201}{0.1404} \approx -17.236$$

Call: glm(formula = y ~ Class + Sex + Age)

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	2.0438	0.1679	12.171	< 2e-16	***
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Null deviance: 2769.5 on 2200 degrees of freedom

Residual deviance: 2210.1 on 2195 degrees of freedom

AIC: 2222.1

3 Test for individual $H_0 : \beta_j = 0$

Q3. Perform Wald test for Sex at level 0.05 $\longrightarrow$ for one $\beta_j = 0$

Q4. Likelihood ratio test for Class at level 0.05 $\longrightarrow$ for some $\beta's = 0$

```
M = glm(y ~ Class + Sex + Age ,  
        data = data,  
        family = binomial)
```

```
Call: glm(formula = y ~ Class + Sex + Age)
```

```
Coefficients:
```

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	2.0438	0.1679	12.171	< 2e-16	***
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```
Null deviance: 2769.5 on 2200 degrees of freedom
```

```
Residual deviance: 2210.1 on 2195 degrees of freedom
```

```
AIC: 2222.1
```

4 Test for the overall $H_0 : \text{some } \beta's = 0$

Q4. Likelihood ratio test for Class at level 0.05 $\longrightarrow$ for some $\beta's = 0$

```
M = glm(y ~ Class + Sex + Age ,  
        data = data,  
        family = binomial)
```

```
Call: glm(formula = y ~ Class + Sex + Age)  
Null deviance: 2769.5 on 2200 degrees of freedom  
Residual deviance: 2210.1 on 2195 degrees of freedom  
AIC: 2222.1
```

$H_0 : \beta_{\text{Class2nd}} = \beta_{\text{Class3rd}} = \beta_{\text{ClassCrew}} = 0$

```
M.noClass = glm(y ~ Sex + Age ,  
                data = data,  
                family = binomial)
```

```
Call: glm(formula = y ~ Sex + Age)  
Null deviance: 2769.5 on 2200 degrees of freedom  
Residual deviance: 2329.1 on 2198 degrees of freedom  
AIC: 2335.1
```

$\text{deviance(M.noClass)} - \text{deviance(M)} \stackrel{H_0}{\sim} \chi^2(3)$

$2329.1 - 2210.1 = 119$

4 Test for the overall $H_0 : \text{some } \beta's = 0$

Q5. Child A/Adult B, everything else same. Derive their survival odds ratio.

```
M = glm(y ~ Class + Sex + Age ,
        data = data,
        family = binomial)
summary(M)
```

Call: glm(formula = y ~ Class + Sex + Age)

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	2.0438	0.1679	12.171	< 2e-16	***
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SexMale	-2.4201	0.1404	-17.236	< 2e-16	***
AgeChild	1.0615	0.2440	4.350	1.36e-05	***

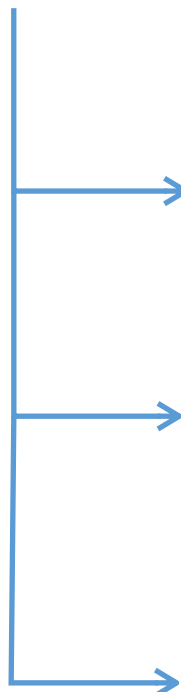
$$\hat{p} = \frac{\exp(2 - x_1 - 1.7x_2 - 0.8x_3 - 2.4x_4 + x_5)}{1 + \exp(2 - x_1 - 1.7x_2 - 0.8x_3 - 2.4x_4 + x_5)}$$

4 Test for the overall $H_0 : \text{some } \beta's = 0$

Q5. Child A/Adult B, everything else same. Derive their survival odds ratio.

$$\hat{p} = \frac{\exp(\star + \hat{\beta}_5 x_5)}{1 + \exp(\star + \hat{\beta}_5 x_5)}$$

$$\hat{p} = \frac{\exp(2 - x_1 - 1.7x_2 - 0.8x_3 - 2.4x_4 + x_5)}{1 + \exp(2 - x_1 - 1.7x_2 - 0.8x_3 - 2.4x_4 + x_5)}$$


$$\hat{p}_{\text{child}} = \frac{\exp(\star + \hat{\beta}_5)}{1 + \exp(\star + \hat{\beta}_5)} \Rightarrow \frac{\hat{p}_{\text{child}}}{1 - \hat{p}_{\text{child}}} = \exp(\star + \hat{\beta}_5)$$

$$\hat{p}_{\text{adult}} = \frac{\exp(\star)}{1 + \exp(\star)} \Rightarrow \frac{\hat{p}_{\text{adult}}}{1 - \hat{p}_{\text{adult}}} = \exp(\star)$$

$$\frac{\text{odds}(\text{child})}{\text{odds}(\text{adult})} = \frac{\exp(\star + \hat{\beta}_5)}{\exp(\star)} = e^{\hat{\beta}_5} = e^{1.06} \approx 2.89.$$

- 1 What is logistic regression?
- 2 Compute an estimate $\hat{\beta}$
- 3 Test for individual $H_0 : \beta_j = 0$
- 4 Test for the overall $H_0 : \text{some } \beta's = 0$
- 5 Given new $\mathbf{x}$, predict its corresponding $\hat{y} = 0/1$

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Q6(a). Predict the survival probability in this dataset.

```
pred.prob = predict(M,  
  newdata = data  
  type = "response")
```

```
> print(pred.prob)  
      1      2      3      4      5  
0.2512 0.2512 0.2512 0.2512 0.2512
```

→ if we don't specify type: it returns log-odds

5 Given new $\mathbf{x}$, predict its corresponding $\hat{y} = 0/1$

Q6(b). Set $\hat{y} = 1$ if $\hat{p} > 0.5$, ow $\hat{y} = 0$. How is the type I/II error rate?

```
pred.prob = predict(M,  
  newdata = data  
  type = "response")
```

```
pred.label = ifelse(pred.prob>0.5, 1, 0)
```

compare

```
conf.mat1 = table(data$y, pred.label)
```

```
type1 = conf.mat1[1,2]/sum(conf.mat1[1, ])
```

```
type2 = conf.mat1[2,1]/sum(conf.mat1[2, ])
```

```
> print(pred.prob)
```

1	2	3	4	5
<u>0.2512</u>	0.2512	0.2512	0.2512	0.2512

0	0	0	0	0
---	---	---	---	---

```
> print(conf.mat1)
```

	pred.label	
	0	1
0	1364	126
1	362	349

5 Given new $\mathbf{x}$, predict its corresponding $\hat{y} = 0/1$

Q7. Set $\hat{y} = 1$ if $\hat{p} > 0.323$, ow $\hat{y} = 0$. How is the type I/II error rate?

```
pred.prob = predict(M,  
  newdata = data  
  type = "response")
```

```
pred.label = ifelse(pred.prob>0.5, 1, 0)
```

```
> print(pred.prob)
```

1	2	3	4	5
0.2512	0.2512	0.2512	0.2512	0.2512

```
conf.mat1 = table(data$y, pred.label)
```

```
type1 = conf.mat1[1,2]/sum(conf.mat1[1, ])
```

```
type2 = conf.mat1[2,1]/sum(conf.mat1[2, ])
```

```
> print(conf.mat1)
```

	pred.label	
	0	1
0	1364	126
1	362	349

Rework

5 Given new $\mathbf{x}$, predict its corresponding $\hat{y} = 0/1$

Q6. Set $\hat{y} = 1$ if $\hat{p} > 0.5$, ow $\hat{y} = 0$. How is the type I/II error rate?

Q7. Set $\hat{y} = 1$ if $\hat{p} > 0.323$, ow $\hat{y} = 0$. How is the type I/II error rate?

Q8. Which threshold is more suitable and why?

Q9. As $0.5 \rightarrow 0.323$, why Type I error increase while Type II error drops?

- 1 What is logistic regression? $p_i = \frac{\exp(\mathbf{x}_i' \boldsymbol{\beta})}{1 + \exp(\mathbf{x}_i' \boldsymbol{\beta})}$
- 2 Compute an estimate $\hat{\boldsymbol{\beta}}$

```
M = glm(y ~ Class + Sex + Age ,  
        data = data,  
        family = binomial)
```
- 3 Test for individual $H_0 : \beta_j = 0$
- 4 Test for the overall $H_0 : \text{some } \beta's = 0$ Deviance difference $\sim \chi^2$
- 5 Given new $\mathbf{x}$, predict its corresponding $\hat{y} = 0/1$ `predict(...)`